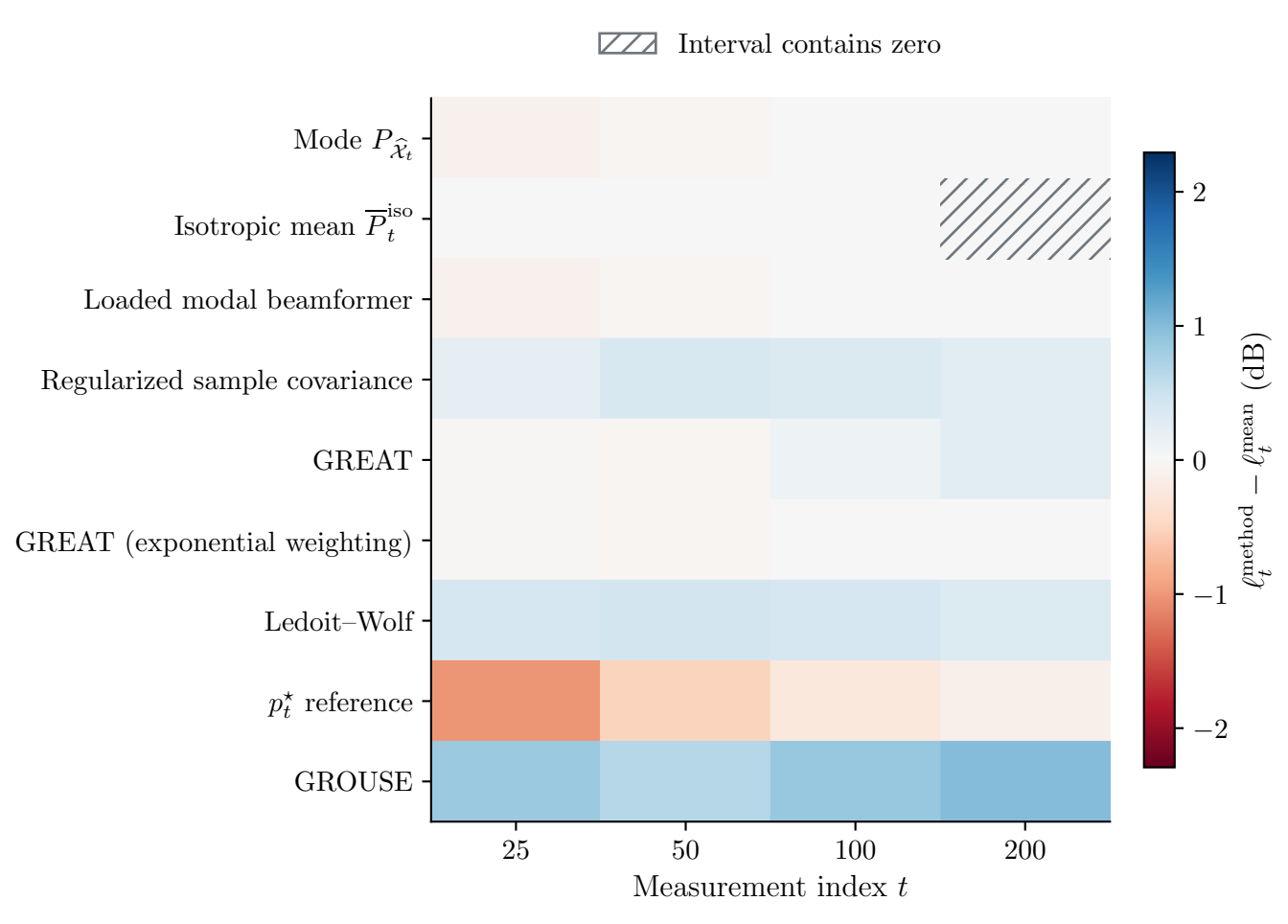
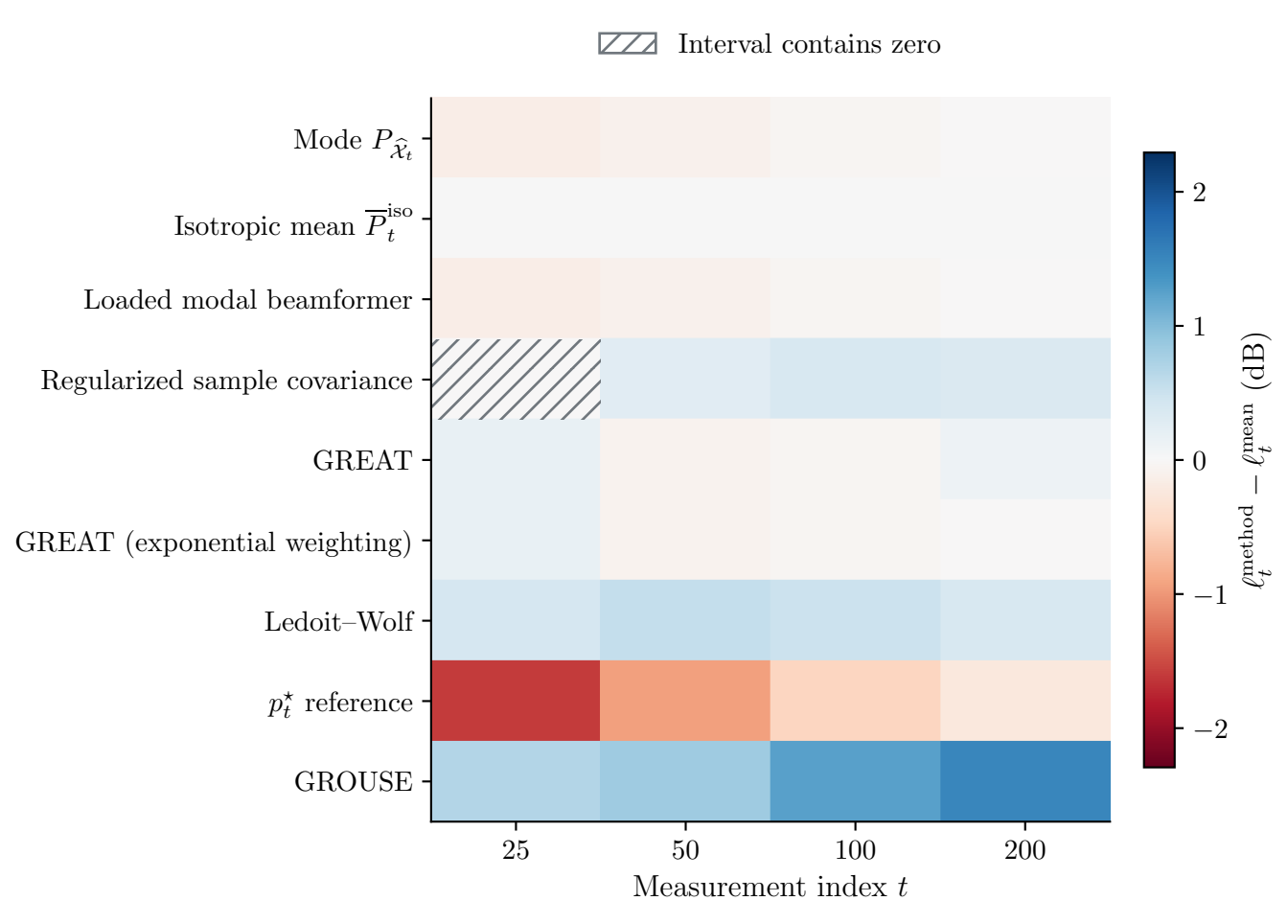
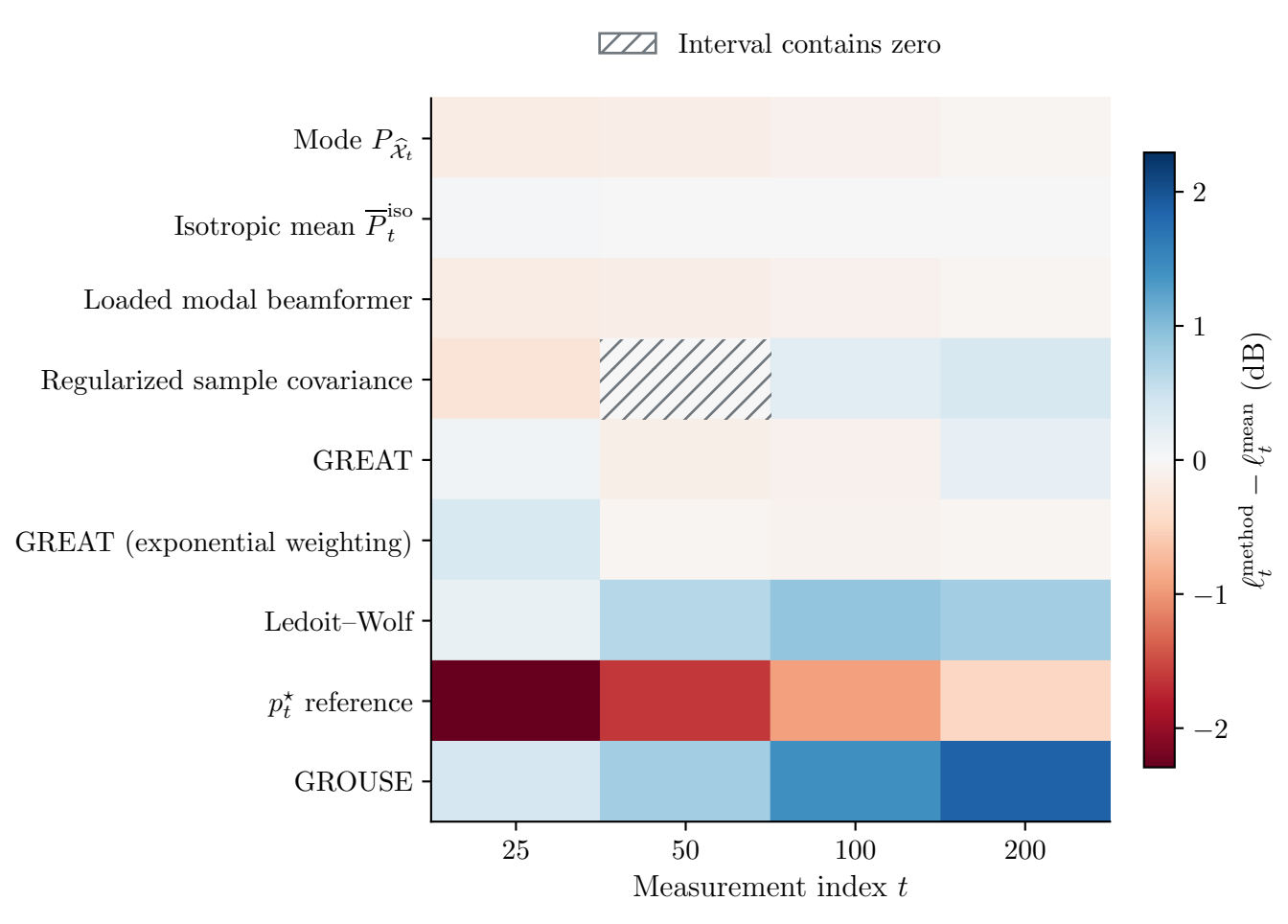
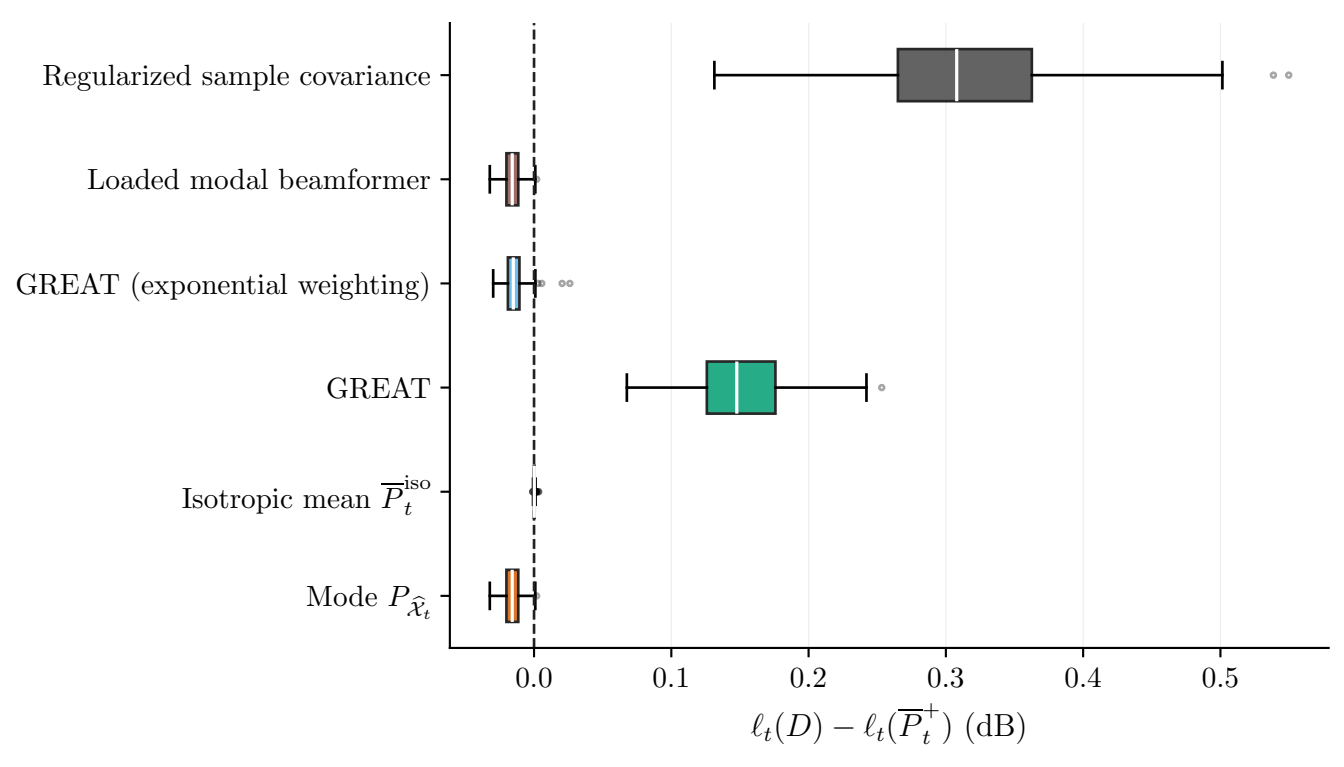
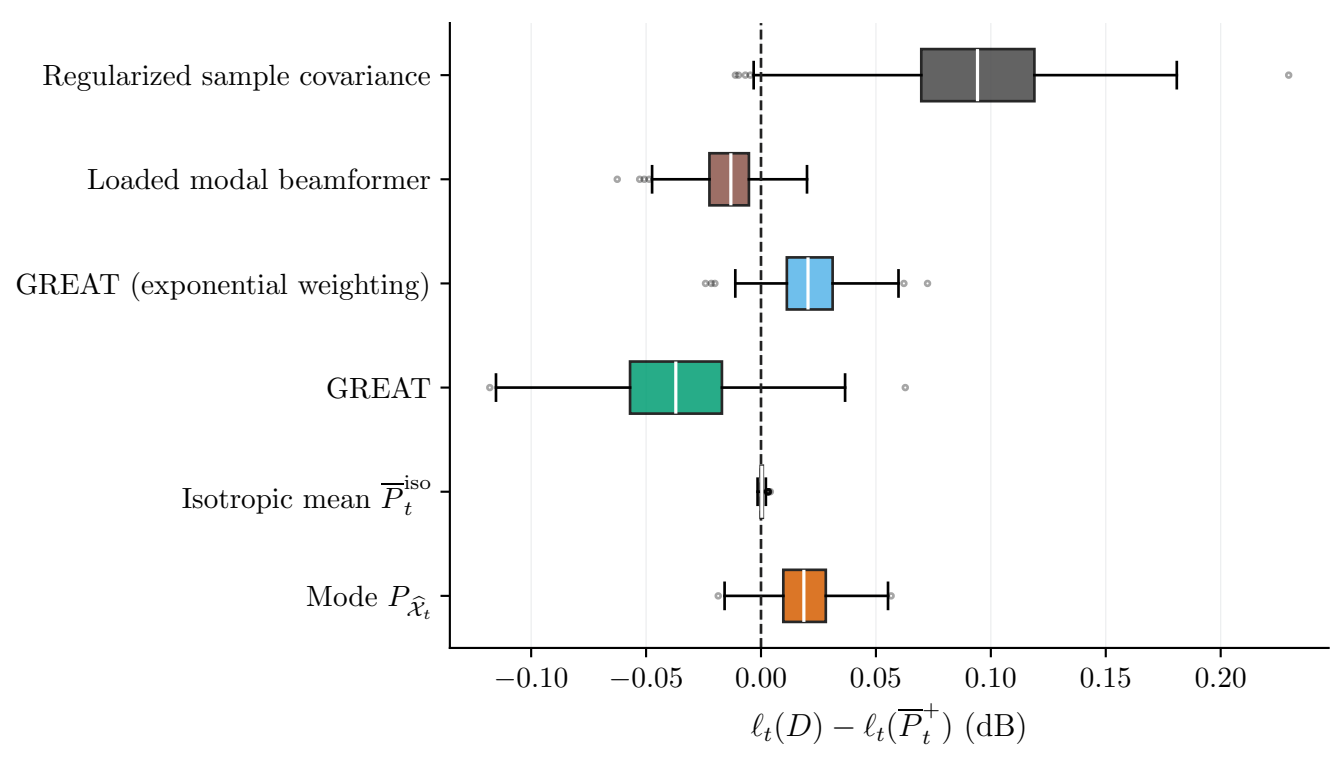
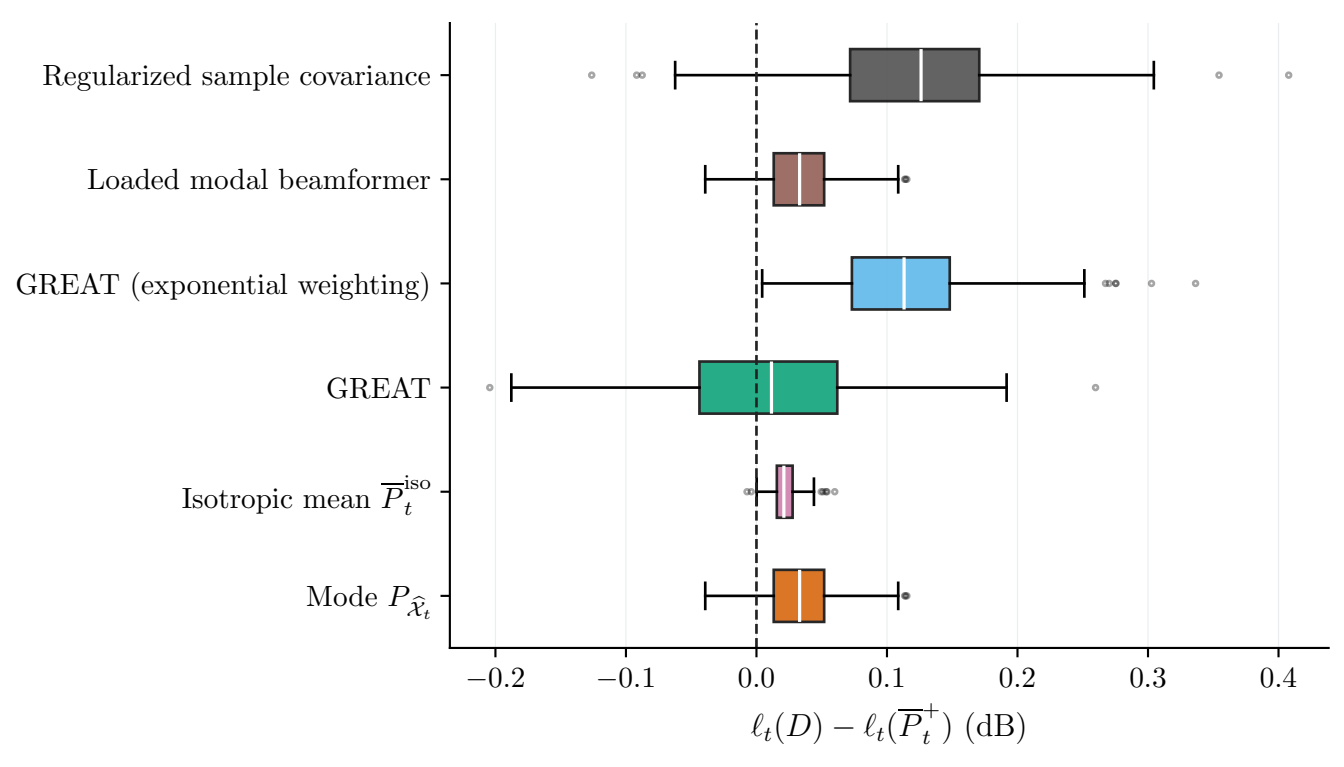

Interval contains zero
Mode $P_{\widehat{\mathcal{X}}_t}$
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Loaded modal beamformer
Regularized sample covariance
GREAT
GREAT (exponential weighting)
Ledoit–Wolf
$p_t^\star$ reference
GROUSE
25
50
100
200
Measurement index t
2
1
0
−1
−2
$\ell_t^{\mathrm{method}} - \ell_t^{\mathrm{mean}}$ (dB)

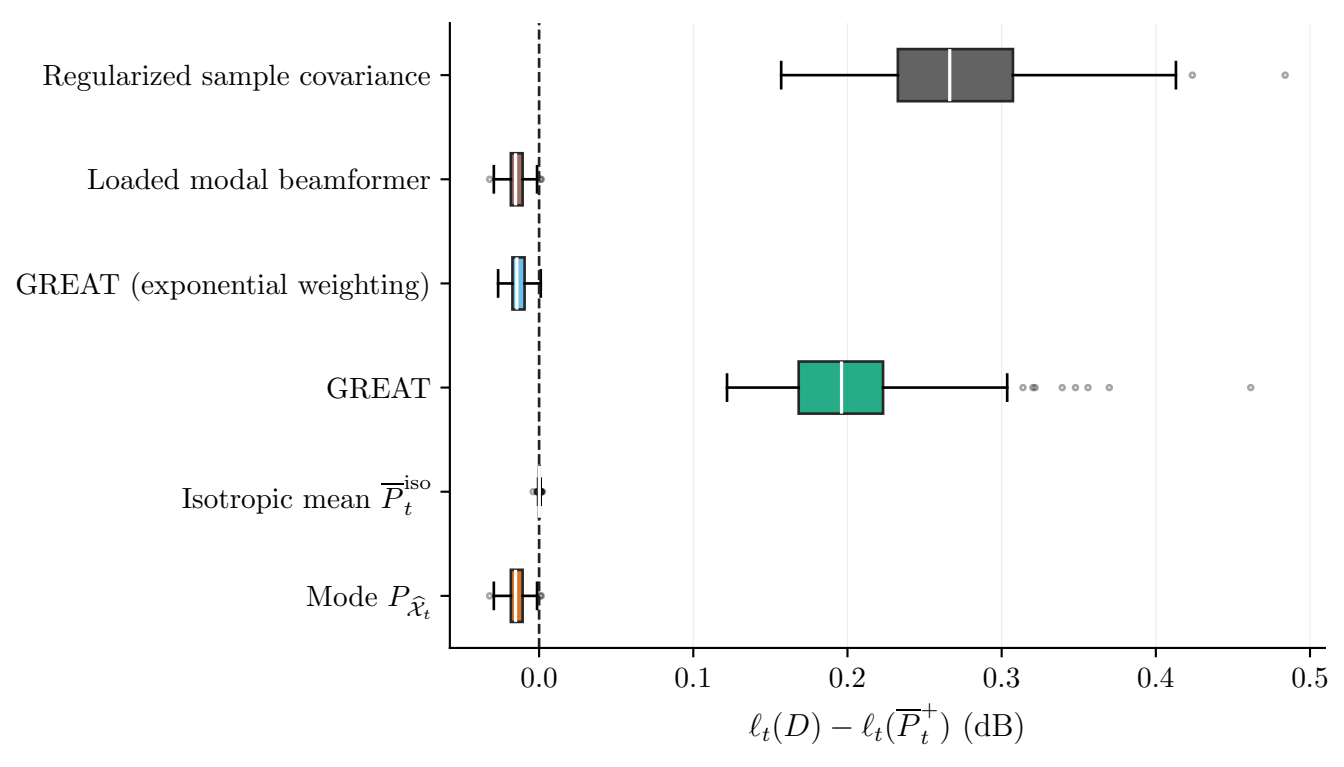

Interval contains zero
Mode $P_{\widehat{\mathcal{X}}_t}$
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Loaded modal beamformer
Regularized sample covariance
GREAT
GREAT (exponential weighting)
Ledoit–Wolf
$p_t^\star$ reference
GROUSE
25
50
100
200
Measurement index t
2
1
0
−1
−2
$\ell_t^{\mathrm{method}} - \ell_t^{\mathrm{mean}}$ (dB)

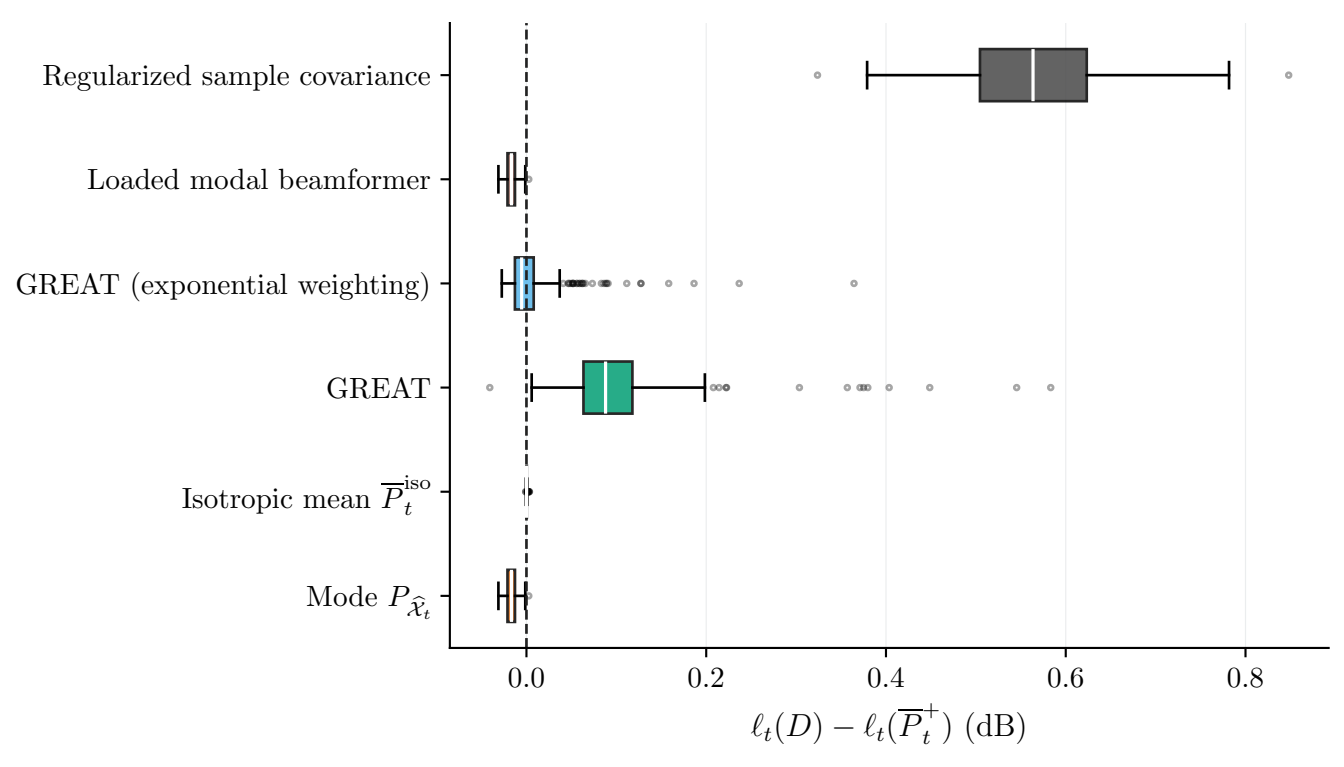

Interval contains zero
Mode $P_{\widehat{\mathcal{X}}_t}$
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Loaded modal beamformer
Regularized sample covariance
GREAT
GREAT (exponential weighting)
Ledoit–Wolf
$p_t^\star$ reference
GROUSE
25
50
100
200
Measurement index t
2
1
0
−1
−2
$\ell_t^{\mathrm{method}} - \ell_t^{\mathrm{mean}}$ (dB)


Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)


Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)


Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)


Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)


Regularized sample covariance
Loaded modal beamformer
GREAT (exponential weighting)
GREAT
Isotropic mean $\overline{P}_t^{\mathrm{iso}}$
Mode $P_{\widehat{\mathcal{X}}_t}$
0.0
0.2
0.4
0.6
0.8
$\ell_t(D) - \ell_t(\overline{P}_t^{+})$ (dB)